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$$\begin{aligned}
 1. \int_0^{\frac{\pi}{2}} \cos^2 \frac{x}{2} dx &= \int_0^{\frac{\pi}{2}} \frac{1 + \cos x}{2} dx \quad (\because \text{반각공식}) \\
 &= \frac{1}{2} [x + \sin x]_0^{\frac{\pi}{2}} \\
 &= \frac{1}{2} \left(\frac{\pi}{2} + 1 \right)
 \end{aligned}$$

$$\begin{aligned}
 2. \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \cot^2 x dx &= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \csc^2 x - 1 dx \\
 &= [-\cot x - x]_{\frac{\pi}{6}}^{\frac{\pi}{4}} \\
 &= \sqrt{3} - 1 - \frac{\pi}{12}
 \end{aligned}$$

$$\begin{aligned}
 3. \int_0^1 \frac{x^4}{x^2+1} dx &= \int_0^1 x^2 - 1 + \frac{1}{x^2+1} dx \\
 &= \left[\frac{1}{3}x^3 - x + \tan^{-1}x \right]_0^1 \\
 &= \frac{\pi}{4} - \frac{2}{3}
 \end{aligned}$$

$$\begin{aligned}
 4. \int_{-1}^1 (e^{x^3} - e^{-x^3}) dx &= \int_{-1}^1 2 \cdot \frac{e^{x^3} - e^{-x^3}}{2} dx = 2 \int_{-1}^1 \sinh(x^3) dx \text{이고} \\
 \sinh x^3 \text{은 기함수이므로 } 2 \int_{-1}^1 \sinh(x^3) dx &= 0 \text{이다.}
 \end{aligned}$$

5. $\tan x$ 을 $\tanh x$ 으로 수정

$$\int_{-2}^2 \left(3 + \sqrt{4-x^2} + \frac{\tanh x}{1+x^2} \right) dx$$

$$1) \int_{-2}^2 3 dx = 12$$

$$2) \int_{-2}^2 \sqrt{4-x^2} dx = 2\pi \quad (\because \text{반원의 넓이})$$

$$3) \frac{\tanh x}{1+x^2} \text{은 기함수이므로 } \int_{-2}^2 \frac{\tanh x}{1+x^2} dx = 0 \text{이다.}$$

1), 2), 3)에 의하여

$$\int_{-2}^2 \left(3 + \sqrt{4-x^2} + \frac{\tanh x}{1+x^2} \right) dx = 12 + 2\pi$$

$$\begin{aligned}
 6. \int_0^1 \frac{2 + \pi x e^{x^2} (x^2 + 1)}{2(x^2 + 1)} dx \\
 &= \int_0^1 \frac{1}{1+x^2} + \frac{\pi}{2} x e^{x^2} dx \\
 &= \left[\tan^{-1}x + \frac{\pi}{4} e^{x^2} \right]_0^1 = \frac{\pi}{4} e
 \end{aligned}$$

7.

$$\begin{aligned}
 &\int_0^{\frac{\pi}{3}} \sec^{2021} x \tan x dx \\
 &= \int_0^{\frac{\pi}{3}} \sec^{2020} x \sec x \tan x dx \\
 &= \frac{1}{2021} [\sec^{2021} x]_0^{\frac{\pi}{3}} \quad (\because \text{전체적분}) \\
 &= \frac{2^{2021} - 1}{2021}
 \end{aligned}$$

8.

$$\begin{aligned}
 \int_1^e \frac{(\ln x)^2}{x(1+(\ln x)^3)} dx &= \frac{1}{3} [\ln(1+(\ln x)^3)]_1^e \quad (\because \text{전체적분}) \\
 &= \frac{1}{3} \ln 2
 \end{aligned}$$

9.

$$\begin{aligned}
 \int_0^{\frac{\pi}{4}} \frac{2e^{\tan x}}{1+\cos 2x} dx &= \int_0^{\frac{\pi}{4}} \frac{e^{\tan x}}{\frac{1+\cos 2x}{2}} dx \\
 &= \int_0^{\frac{\pi}{4}} \frac{e^{\tan x}}{\cos^2 x} dx \\
 &= \int_0^{\frac{\pi}{4}} e^{\tan x} \sec^2 x dx \\
 &= [e^{\tan x}]_0^{\frac{\pi}{4}} = e - 1 \quad (\because \text{전체적분})
 \end{aligned}$$

10.

$$\begin{aligned}
 &\int_0^1 \frac{\sqrt{\cosh x + 1}}{\sqrt[3]{\cosh x - 1}} dx \\
 &= \int_0^1 \frac{\sqrt{\cosh x + 1}}{\sqrt[3]{\cosh x - 1}} \times \frac{\sqrt{\cosh x - 1}}{\sqrt{\cosh x - 1}} dx \\
 &= \int_0^1 \frac{\sqrt{\cosh^2 x - 1}}{(\cosh x - 1)^{\frac{5}{6}}} dx \\
 &= \int_0^1 \frac{\sinh x}{(\cosh x - 1)^{\frac{5}{6}}} dx \\
 &= 6 \left[(\cosh x - 1)^{\frac{1}{6}} \right]_0^1 \quad (\because \text{전체적분}) \\
 &= 6 \sqrt[6]{\cosh 1 - 1}
 \end{aligned}$$

11.

$$\begin{aligned}
 &\int_0^{\frac{\pi}{6}} \frac{\sin x \cos x}{\sin^4 x + \cos^4 x} dx \\
 &= \int_0^{\frac{\pi}{6}} \frac{\tan x \sec^2 x}{1 + \tan^4 x} dx \quad (\tan x = t \text{로 치환}) \\
 &= \int_0^{\frac{1}{\sqrt{3}}} \frac{t}{1+t^4} dt \\
 &= \int_0^{\frac{1}{\sqrt{3}}} \frac{t}{1+(t^2)^2} dt \\
 &= \frac{1}{2} [\tan^{-1}(t^2)]_0^{\frac{1}{\sqrt{3}}} \\
 &= \frac{1}{2} \tan^{-1}\left(\frac{1}{3}\right) = \alpha \Rightarrow \tan(2\alpha) = \frac{1}{3}, \therefore \sin 2\alpha = \frac{1}{\sqrt{10}}
 \end{aligned}$$

12. $\arccos x = \alpha$ 으로 치환하면 $\cos \alpha = x, \sin \alpha = \sqrt{1-x^2}$ 이므로

$$\begin{aligned} & \int_0^1 \sin(2\arccos x) dx \\ &= \int_0^1 \sin(2\alpha) dx \\ &= \int_0^1 2\sin\alpha\cos\alpha dx \\ &= \int_0^1 2x\sqrt{1-x^2} dx \\ &= -\int_0^1 -2x(1-x^2)^{\frac{1}{2}} dx \\ &= -\frac{2}{3} \left[(1-x^2)^{\frac{3}{2}} \right]_0^1 = \frac{2}{3} \end{aligned}$$

13.

$$\begin{aligned} & \int_1^2 \frac{\sqrt{x^2-1}}{x} dx \\ &= \int_0^{\frac{\pi}{3}} \frac{\tan\theta}{\sec\theta} \sec\theta \tan\theta d\theta \quad (\because x = \sec\theta) \\ &= \int_0^{\frac{\pi}{3}} \tan^2\theta d\theta \\ &= \int_0^{\frac{\pi}{3}} \sec^2\theta - 1 d\theta \\ &= [\tan\theta - \theta]_0^{\frac{\pi}{3}} = \sqrt{3} - \frac{\pi}{3} \end{aligned}$$

14.

$x = \frac{3}{2} \sin\theta$ 로 삼각치환하면 $dx = \frac{3}{2} \cos\theta d\theta$ 이므로
주어진 식은 다음과 같다.

$$\begin{aligned} & \int_0^{\frac{3}{2}} \frac{x^2}{\sqrt{9-4x^2}} dx \\ &= \int_0^{\frac{\pi}{2}} \frac{\frac{9}{4} \sin^2\theta}{\sqrt{9-4\left(\frac{9}{4} \sin^2\theta\right)}} \cdot \frac{3}{2} \cos\theta d\theta \\ &= \frac{9}{8} \int_0^{\frac{\pi}{2}} \sin^2\theta d\theta \\ &= \frac{9}{16} \int_0^{\frac{\pi}{2}} 1 - \cos 2\theta d\theta \\ &= \frac{9}{16} \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{2}} = \frac{9\pi}{32} \end{aligned}$$

[다른 풀이]

$$\begin{aligned} \frac{9}{8} \int_0^{\frac{\pi}{2}} \sin^2\theta d\theta &= \frac{9}{8} \times \left(\frac{1}{2} \cdot \frac{\pi}{2} \right) \quad (\because \text{wallis 공식}) \\ &= \frac{9\pi}{32} \end{aligned}$$

15.

$x = \sqrt{2} \tan\theta$ 로 치환하면

$$\begin{aligned} & \int_0^{\sqrt{2}} x^3 \sqrt{x^2+2} dx \\ &= \int_0^{\frac{\pi}{4}} 2\sqrt{2} \tan^3\theta \sqrt{2} \sec\theta \sqrt{2} \sec^2\theta d\theta \\ &= 4\sqrt{2} \int_0^{\frac{\pi}{4}} \tan^3\theta \sec^3\theta d\theta \\ &= 4\sqrt{2} \int_0^{\frac{\pi}{4}} \tan^2\theta \sec^2\theta \sec\theta \tan\theta d\theta \quad (\sec\theta = t) \\ &= 4\sqrt{2} \int_1^{\sqrt{2}} (t^2-1)t^2 dt \\ &= 4\sqrt{2} \int_1^{\sqrt{2}} (t^4 - t^2) dt \end{aligned}$$

$$\begin{aligned} &= 4\sqrt{2} \left[\frac{1}{5} t^5 - \frac{1}{3} t^3 \right]_1^{\sqrt{2}} \\ &= \frac{8}{15} (2 + \sqrt{2}) \end{aligned}$$

[다른 풀이]

$x^2 + 2 = t$ 로 치환하면 $x dx = \frac{1}{2} dt$ 이므로

$$\begin{aligned} & \int_0^{\sqrt{2}} x^3 \sqrt{x^2+2} dx = \int_2^4 \frac{1}{2} (t-2) \sqrt{t} dt \\ &= \frac{1}{2} \int_2^4 t^{\frac{3}{2}} - 2t^{\frac{1}{2}} dt \\ &= \frac{1}{2} \left[\frac{2}{5} t^{\frac{5}{2}} - \frac{4}{3} t^{\frac{3}{2}} \right]_2^4 \\ &= \frac{1}{2} \left(\left(\frac{64}{5} - \frac{32}{3} \right) - \left(\frac{8}{5} \sqrt{2} - \frac{8}{3} \sqrt{2} \right) \right) = \frac{8}{15} (2 + \sqrt{2}) \text{이다.} \end{aligned}$$

16.

$$\begin{aligned} & \int_0^1 x^5 e^{x^2} dx \quad (x^2 = t \text{로 치환}) \\ &= \int_0^1 \frac{1}{2} t^2 e^t dt \\ &= \frac{1}{2} [t^2 e^t - 2te^t + 2e^t]_0^1 \quad (\because \text{빠른 부분적분}) \\ &= \frac{1}{2} (e - 2e + 2e - 2) \\ &= \frac{1}{2} (e - 2) = \frac{1}{2} e - 1 \end{aligned}$$

17.

$$\begin{aligned} & \int_{\frac{1}{\sqrt{2}}}^1 \cos^{-1} x dx \\ &= \left[x \cos^{-1} x \right]_{\frac{1}{\sqrt{2}}}^1 - \int_{\frac{1}{\sqrt{2}}}^1 x \frac{-1}{\sqrt{1-x^2}} dx \quad (\because \text{부분적분}) \\ &= -\frac{\sqrt{2}\pi}{8} + \int_{\frac{1}{\sqrt{2}}}^1 x(1-x^2)^{-\frac{1}{2}} dx \\ &= -\frac{\sqrt{2}\pi}{8} - \left[(1-x^2)^{\frac{1}{2}} \right]_{\frac{1}{\sqrt{2}}}^1 \quad (\because \text{전체적분}) \\ &= \frac{\sqrt{2}(4-\pi)}{8} \end{aligned}$$

18.

$$\begin{aligned} & \int_0^1 \ln(x + \sqrt{x^2+1}) dx \\ &= \int_0^1 \sinh^{-1} x dx \\ &= [x \sinh^{-1} x]_0^1 - \int_0^1 \frac{x}{\sqrt{1+x^2}} dx \quad (\because \text{부분적분}) \\ &= \ln(1 + \sqrt{2}) - [\sqrt{1+x^2}]_0^1 \quad (\because \text{전체적분}) \\ &= \ln(1 + \sqrt{2}) - \{\sqrt{2} - 1\} \\ &= 1 - \sqrt{2} + \ln(1 + \sqrt{2}) \end{aligned}$$

19.

$$\begin{aligned} & \int_1^2 x \arcsin\left(\frac{1}{x}\right) dx \\ &= \left[\frac{1}{2} x^2 \sin^{-1}\left(\frac{1}{x}\right) \right]_1^2 - \int_1^2 \frac{1}{2} x^2 \frac{1}{\sqrt{1-\left(\frac{1}{x}\right)^2}} \left(-\frac{1}{x^2}\right) dx \quad (\because \text{부분적분}) \\ &= \frac{\pi}{12} + \int_1^2 \frac{1}{2} \frac{x}{\sqrt{x^2-1}} dx \\ &= \frac{\pi}{12} + \frac{1}{2} \left[(x^2-1)^{\frac{1}{2}} \right]_1^2 \quad (\because \text{전체적분}) \\ &= \frac{\pi}{12} + \frac{\sqrt{3}}{2} \end{aligned}$$

20.

$$\begin{aligned}
 & \int_{\frac{1}{2}}^1 (\arccos x)^2 dx \quad (\cos^{-1}x = t \text{로 치환}) \\
 &= \int_{\frac{\pi}{3}}^0 t^2 (-\sin t) dt \\
 &= \int_0^{\frac{\pi}{3}} t^2 \sin t dt \\
 &= [t^2(-\cos t) - 2t(-\sin t) + 2\cos t]_0^{\frac{\pi}{3}} \quad (\because \text{빠른 부분적분}) \\
 &= [-t^2 \cos t + 2t \sin t + 2\cos t]_0^{\frac{\pi}{3}} \\
 &= -\frac{\pi^2}{9} \cdot \frac{1}{2} + 2 \cdot \frac{\pi}{3} \cdot \frac{\sqrt{3}}{2} + 2 \times \frac{1}{2} - 2 \\
 &= -\frac{\pi^2}{18} + \frac{\sqrt{3}}{3} \pi - 1
 \end{aligned}$$

21.

$$\begin{aligned}
 & \int_1^e \frac{(\ln x)^2}{x^2} dx \quad (\ln x = t \text{로 치환}) \\
 &= \int_0^1 t^2 e^{-t} dt \\
 &= [-t^2 e^{-t} - 2te^{-t} - 2e^{-t}]_0^1 \quad (\because \text{빠른 부분적분}) \\
 &= 2 - 5e^{-1}
 \end{aligned}$$

22.

$$\begin{aligned}
 & \int_1^e (\ln x)^3 dx \quad (\ln x = t \text{로 치환}) \\
 &= \int_0^1 t^3 e^t dt \\
 &= [t^3 e^t - 3t^2 e^t + 6te^t - 6e^t]_0^1 \quad (\because \text{빠른 부분적분}) \\
 &= 6 - 2e
 \end{aligned}$$

23.

$$\begin{aligned}
 & \int_0^\infty x e^{-x^2} \cos(2x^2) dx \quad (x^2 = t) \\
 &= \frac{1}{2} \int_0^\infty e^{-t} \cos(2t) dt \\
 &= \frac{1}{2} \left[\frac{e^{-t}}{1+2^2} (-1 \cos 2t + 2 \sin 2t) \right]_0^\infty \\
 &= \frac{1}{10}
 \end{aligned}$$

24.

$$\begin{aligned}
 & \int_1^\pi \left(\frac{\cos x}{x} - \frac{3}{2} \frac{\sin x}{x^2} \right) \cdot \frac{1}{\sqrt{x}} dx \\
 &= \int_1^\pi x^{-\frac{3}{2}} \cos x - \frac{3}{2} x^{-\frac{5}{2}} \sin x dx \\
 &= \left[x^{-\frac{3}{2}} \sin x \right]_1^\pi \quad (\because \text{곱미분}) \\
 &= -\sin 1 \\
 & \text{[다른 풀이]} \\
 & \int_1^\pi \left(\frac{\cos x}{x} - \frac{3}{2} \frac{\sin x}{x^2} \right) \cdot \frac{1}{\sqrt{x}} dx \\
 &= \int_1^\pi x^{-\frac{3}{2}} \cos x - \frac{3}{2} x^{-\frac{5}{2}} \sin x dx \\
 &= \left[x^{-\frac{3}{2}} \sin x \right]_1^\pi - \int_1^\pi -\frac{3}{2} x^{-\frac{5}{2}} \sin x dx + \int_1^\pi -\frac{3}{2} x^{-\frac{5}{2}} \sin x dx \\
 &= -\sin 1
 \end{aligned}$$

25.

$$\begin{aligned}
 & \int_0^1 \left(2x \sin \frac{1}{x^2} - \frac{2}{x} \cos \frac{1}{x^2} \right) dx \\
 &= \left[x^2 \sin \left(\frac{1}{x^2} \right) \right]_0^1
 \end{aligned}$$

= sin 1

[다른 풀이]

$$\begin{aligned}
 & \int_0^1 \left(2x \sin \left(\frac{1}{x^2} \right) - \frac{2}{x} \cos \left(\frac{1}{x^2} \right) \right) dx \\
 &= \int_0^1 2x \sin \left(\frac{1}{x^2} \right) dx - \int_0^1 \frac{2}{x} \cos \left(\frac{1}{x^2} \right) dx \\
 &= \left[x^2 \sin \frac{1}{x^2} \right]_0^1 - \int_0^1 x^2 \cos \frac{1}{x^2} \left(-\frac{2}{x^3} \right) dx - \int_0^1 \frac{2}{x} \cos \left(\frac{1}{x^2} \right) dx \\
 &= \sin 1 + \int_0^1 \frac{2}{x} \cos \frac{1}{x^2} dx - \int_0^1 \frac{2}{x} \cos \left(\frac{1}{x^2} \right) dx \\
 &= \sin 1
 \end{aligned}$$

26.

$$\begin{aligned}
 & \int_1^2 \frac{3x+1}{x^2+x} dx \\
 &= \int_1^2 \frac{3x+1}{x(x+1)} dx \\
 &= \int_1^2 \frac{1}{x} + \frac{2}{x+1} dx \\
 &= [\ln x + 2 \ln(x+1)]_1^2 \\
 &= \ln \left(\frac{9}{2} \right)
 \end{aligned}$$

27.

$$\begin{aligned}
 & \int_3^4 \frac{1}{x^2-4} dx \\
 &= \int_3^4 \frac{1}{(x+2)(x-2)} dx \\
 &= \int_3^4 \frac{-\frac{1}{4}}{x+2} + \frac{\frac{1}{4}}{x-2} dx \\
 &= \frac{1}{4} [-\ln(x+2) + \ln(x-2)]_3^4 \\
 &= \frac{1}{4} \ln \frac{5}{3}
 \end{aligned}$$

$$\text{참고. } \int \frac{1}{x^2-a^2} dx = \frac{1}{2a} \ln \left(\frac{x-a}{x+a} \right) + C$$

[다른 풀이]

$$\int_3^4 \frac{1}{x^2-4} dx = \frac{1}{4} \left[\ln \left(\frac{x-2}{x+2} \right) \right]_3^4 = \frac{1}{4} \ln \frac{5}{3}$$

28.

$$\begin{aligned}
 & \int_2^3 \frac{x^2+1}{(x+1)(x-1)^2} dx \\
 &= \int_2^3 \frac{\frac{1}{2}}{x+1} + \frac{\frac{1}{2}}{x-1} + \frac{1}{(x-1)^2} dx \\
 &= \left[\frac{1}{2} \ln(x+1) + \frac{1}{2} \ln(x-1) - \frac{1}{(x-1)} \right]_2^3 \\
 &= \frac{1}{2} (3 \ln 2 - \ln 3 + 1)
 \end{aligned}$$

29.

$$\begin{aligned}
 & \int \frac{f(x)}{x^3(x+1)^2} dx \\
 &= \int \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{D}{x+1} + \frac{E}{(x+1)^2} dx \\
 & \text{여기서 유리함수를 만족하려면 } A=D=0 \text{이다.} \\
 & \int \frac{B}{x^2} + \frac{C}{x^3} + \frac{E}{(x+1)^2} dx \\
 &= \int \frac{Bx(x+1)^2 + C(x+1)^2 + Ex^3}{x^3(x+1)^2} dx
 \end{aligned}$$

따라서 $f(x) = Bx(x+1)^2 + C(x+1)^2 + Ex^3$ 이다.

$f(x)$ 는 이차함수이고, $f(0)=1$, $f(1)=10$ 을 만족하므로

$B+E=0$, $C=1$, $4B+4C+E=10$ 이므로

$B=2$, $E=-2$, $C=1$ 이다.

즉, $f(x) = 5x^2 + 4x + 1$ 이므로 $f(2) = 29$ 이다.

30.

$$\begin{aligned}
& \int_0^1 \frac{4x^2 + 17x + 1}{(x+2)(4x^2+1)} dx \\
&= \int_0^1 \frac{-1}{x+2} + \frac{8x+1}{4x^2+1} dx \\
&= \int_0^1 \frac{-1}{x+2} + \frac{8x}{4x^2+1} + \frac{1}{1+4x^2} dx \\
&= \left[-\ln(x+2) + \ln(4x^2+1) + \frac{1}{2} \tan^{-1}(2x) \right]_0^1 \\
&= \ln \frac{10}{3} + \frac{1}{2} \tan^{-1} 2
\end{aligned}$$

31.

$$\begin{aligned}
& \int_0^1 \frac{1}{(x^2+1)(x^2+4)} dx \\
&= \int_0^1 \frac{\frac{1}{3}}{x^2+1} - \frac{\frac{1}{3}}{x^2+4} dx \\
&= \frac{1}{3} \left[\tan^{-1} x - \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) \right]_0^1 \\
&= \frac{1}{3} \left(\frac{\pi}{4} - \frac{1}{2} \tan^{-1} \frac{1}{2} \right)
\end{aligned}$$

32.

$$\begin{aligned}
& \lim_{t \rightarrow \infty} f(t) \\
&= 2\pi \int_0^\infty \frac{x^2}{(x^2+1)(x^2+4)(x^2+9)} dx \\
&= 2\pi \int_0^\infty \frac{-\frac{1}{24}}{x^2+1} + \frac{\frac{4}{15}}{x^2+4} - \frac{\frac{9}{40}}{x^2+9} dx \\
&= 2\pi \left[-\frac{1}{24} \tan^{-1} x + \frac{4}{15} \cdot \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) - \frac{9}{40} \cdot \frac{1}{3} \tan^{-1} \left(\frac{x}{3} \right) \right]_0^\infty \\
&= \frac{\pi^2}{60}
\end{aligned}$$

33.

$$\begin{aligned}
& \int_0^1 \frac{x+7}{x^2+4x+5} dx \\
&= \int_0^1 \frac{\frac{1}{2}(2x+4)+5}{x^2+4x+5} dx \\
&= \int_0^1 \frac{\frac{1}{2}(2x+4)}{x^2+4x+5} + \frac{5}{x^2+4x+5} dx \\
&= \int_0^1 \frac{\frac{1}{2}(2x+4)}{x^2+4x+5} + \frac{5}{(x+2)^2+1} dx \\
&= \left[\frac{1}{2} \ln(x^2+4x+5) + 5 \tan^{-1}(x+2) \right]_0^1 \\
&= \frac{1}{2} \ln 2 + 5(\tan^{-1} 3 - \tan^{-1} 2) \\
&= \ln \sqrt{2} + 5 \tan^{-1} \frac{1}{7}
\end{aligned}$$

34.

$$\begin{aligned}
& \sqrt{x} = t \text{로 치환하면 } x = t^2, dx = 2t dt \text{ 이고} \\
& \int_1^4 \frac{1}{x + \sqrt{x}} dx = \int_1^2 \frac{1}{t^2 + t} \cdot 2t dt = \int_1^2 \frac{2}{t+1} dt \\
&= 2[\ln|t+1|]_1^2 = 2(\ln 3 - \ln 2) = \ln \frac{9}{4}
\end{aligned}$$

35.

$$\begin{aligned}
& \sqrt{1 + \sqrt{2+x}} = t \text{로 치환하면} \\
& 1 + \sqrt{2+x} = t^2, \quad 2+x = t^4 - 2t^2 + 1, \quad dx = (4t^3 - 4t) dt \text{ 이다.} \\
& \int_7^{62} \frac{dx}{\sqrt{1 + \sqrt{2+x}}}
\end{aligned}$$

$$\begin{aligned}
&= \int_2^3 \frac{1}{t} (4t^3 - 4t) dt \\
&= \int_2^3 (4t^2 - 4) dt \\
&= 4 \left[\frac{1}{3} t^3 - t \right]_2^3 \\
&= \frac{64}{3}
\end{aligned}$$

36.

$$\begin{aligned}
& \sqrt[6]{x} = t \text{로 치환하면 } \sqrt{x} = t^3, \sqrt[3]{x} = t^2, x = t^6, dx = 6t^5 dt \text{ 이다.} \\
& \int_0^1 \frac{1}{\sqrt{x} + \sqrt[3]{x}} dx \\
&= \int_0^1 \frac{6t^5}{t^3 + t^2} dt \\
&= 6 \int_0^1 \frac{t^3}{t+1} dt \\
&= 6 \int_0^1 t^2 - t + 1 + \frac{-1}{t+1} dt \\
&= 6 \left[\frac{1}{3} t^3 - \frac{1}{2} t^2 + t - \ln(t+1) \right]_0^1 \\
&= 5 - 6 \ln 2
\end{aligned}$$

37.

$$\begin{aligned}
& \int_1^5 \frac{1}{\sqrt{t^2 - 6t + 13}} dt \\
&= \int_1^5 \frac{1}{\sqrt{(t-3)^2 + 4}} dt \\
&= \left[\sinh^{-1} \left(\frac{t-3}{2} \right) \right]_1^5 \\
&= \sinh^{-1}(1) - \sinh^{-1}(-1) \\
&= 2 \sinh^{-1}(1) \\
&= 2 \ln(1 + \sqrt{2}) \\
&= \ln(3 + 2\sqrt{2})
\end{aligned}$$

38.

$$\begin{aligned}
& e^x = t \text{로 치환하면 } x = \ln t, dx = \frac{1}{t} dt \text{ 이다.} \\
& \int_{\ln 2}^{\ln 3} e^{-x} \coth x dx \\
&= \int_{\ln 2}^{\ln 3} e^{-x} \frac{e^{2x} + 1}{e^{2x} - 1} dx \\
&= \int_2^3 \frac{1}{t} \frac{t^2 + 1}{t^2 - 1} \frac{1}{t} dt \\
&= \int_2^3 \frac{t^2 + 1}{t^2(t^2 - 1)} dt \\
&= \int_2^3 \frac{-1}{t^2} + \frac{2}{t^2 - 1} dt \\
&= \left[\frac{1}{t} + \ln \left(\frac{t-1}{t+1} \right) \right]_2^3 \\
&= \ln \frac{3}{2} - \frac{1}{6}
\end{aligned}$$

39.

$$\begin{aligned}
& \int_0^\pi \frac{1}{5 + 3 \sin x} dx \quad \left(\tan \left(\frac{x}{2} \right) = t \text{로 치환} \right) \\
&= \int_0^\infty \frac{1}{\left(5 + 3 \frac{2t}{1+t^2} \right)} \frac{2}{1+t^2} dt \\
&= \int_0^\infty \frac{2}{5t^2 + 6t + 5} dt \\
&= \frac{2}{5} \int_0^\infty \frac{1}{t^2 + \frac{6}{5}t + 1} dt
\end{aligned}$$

$$\begin{aligned}
&= \frac{2}{5} \int_0^{\infty} \frac{1}{\left(t + \frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} dt \\
&= \frac{2}{5} \frac{1}{\frac{4}{5}} \left[\tan^{-1} \left(\frac{t + \frac{3}{5}}{\frac{4}{5}} \right) \right]_0^{\infty} \\
&= \frac{1}{2} \left(\frac{\pi}{2} - \tan^{-1} \frac{3}{4} \right)
\end{aligned}$$

$$\begin{aligned}
40. \quad &\int_0^x \sin 5\theta \sin 3\theta \, d\theta \\
&= \int_0^x -\frac{1}{2} (\cos(8\theta) - \cos(2\theta)) d\theta \\
&= -\frac{1}{2} \left[\frac{1}{8} \sin 8\theta - \frac{1}{2} \sin 2\theta \right]_0^x \\
&= -\frac{1}{16} \sin 8x + \frac{1}{4} \sin 2x
\end{aligned}$$

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